[TITLE — working: Does random maneuvering help a drone slip past sensors? A diffusion model for counter-UAS sensor layouts]

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Abstract

[ABSTRACT --- about 200 words. Problem; model (diffusing drone, Poisson detection, Feynman--Kac); validation; four findings (Q1--Q4); real-world reading; one design rule.]

1 INTRODUCTION

[Motivation: small-UAS threat; layered counter-UAS; where sensors go matters.]

[Gap: planners and most models assume straight-line or fully informed intruders; the intermediate, randomly maneuvering intruder is unstudied (see Section 2).]

[Questions Q1--Q4, each in one line: Q1 does randomness help the drone (regime map); Q2 does it survive an equal-time comparison; Q3 where to put the sensors; Q4 what sensor downtime does and how much redundancy offsets it.]

[Contributions list (also the separate MOR statement of contribution).]

[Roadmap of the paper.]

2 RELATED WORK

2.1 Search and Detection Theory

[Classical search/detection; detection as a Poisson process (cite).]

2.2 Barrier Coverage and Intrusion Detection

[Sensor-network intrusion detection, e.g. Dousse, Tavoularis and Thiran (2006); barrier coverage.]

2.3 Counter-UAS Modeling

[Kim et al. (2025, planned-evasion counter-UAS); other C-UAS layout studies; Eagle (1987) citing literature.]

2.4 Optimal Placement of Traps and Sensors

[Kolokolnikov, Titcombe and Ward (2005): ring vs ring-plus-center patterns; contrast with our result.]

[Positioning sentence: intruder between the straight-line and the fully informed adversary.]

3 MODEL

3.1 Drone Motion

[SDE with inward drift and diffusion; reflecting outer edge; arrival at the response line r = d.]

3.2 Sensors and Detection

[Gaussian footprints; Poisson detection with rate kappa(x); kernels considered.]

3.3 Chance of Arriving Undetected

[Feynman--Kac PDE with boundary conditions (equation); worst-case entry objective F(s).]

3.4 Dimensionless Numbers

[Pe and Da; response distance d and depth R - d; parameter ranges used.]

[TABLE: symbols, meanings, units, ranges.]

4 NUMERICAL METHODS AND VALIDATION

4.1 Finite-Difference Solver

[Polar grid, conservative diffusion, hybrid drift scheme, ghost ring, sparse direct solve.]

4.2 Monte Carlo Simulator

[Euler--Maruyama, survival weights, Brownian-bridge crossing, half-step refund.]

4.3 Validation Ladder

[Rungs 1--5 and results (max |z| = 1.98; orders 1.99 central, 0.99 upwind).]

[FIGURE: figures/validation.png (re-render in grayscale).]

[TABLE: validation summary from results/validation.md.]

5 DOES RANDOM MANEUVERING HELP THE DRONE?

5.1 Regime Map

[Q1: random vs straight-line worst case over (Pe, Da); where randomness helps/harms; when the straight-line shortcut is accurate.]

[FIGURE: figures/regime_map.png (grayscale).]

[Monte Carlo spot check: results/regime_spotcheck.csv.]

5.2 Equal-Time Comparison

[Q2: matching mean arrival time; results/regime_equal_time.csv.]

[FIGURE: figures/regime_equal_time.png (grayscale).]

5.3 Robustness to Layout and Sensor Gap

[Layouts K4--K8 and ring radii; same-gap collapse; regime_layouts / regime_gap_test.]

[FIGURE: figures/regime_gap_test.png or regime_layouts.png (choose one).]

6 WHERE SHOULD THE SENSORS GO?

6.1 The Best Ring and the Cost of Straight-Line Planning

[Best ring radius by drone type and K; cost of the straight-line choice; response distance.]

[FIGURE: figures/q3_best_ring.png (grayscale).]

[Sensitivity to footprint and rate errors: results/q3_sensitivity.csv.]

6.2 Is the Ring Optimal? Numerical Evidence

[Structured families, adjoint gradient (108 runs), seeded DE, multi-start Powell; comparison table.]

[TABLE: results/q3_layout_comparison.md (condensed).]

[FIGURE: figures/q3_adjoint_search.png (grayscale).]

6.3 Theory

[Midpoint theorem; Clarke stationarity; straight-line results; local optimality (to re-verify); K = 4 branch-and-bound certificate; best radius vs randomness; lower bound (floor). Source: paper/theory_section.tex.]

[TABLE: best ring radius vs D (PDE vs asymptotics).]

7 SENSOR DOWNTIME AND REDUNDANCY

7.1 Markov On/Off Sensors

[Two-state chain; availability p and switching rate gamma; fast and frozen limits; Jensen ordering.]

7.2 Brief Versus Long Outages

[FIGURE: figures/q4_downtime_switching.png (grayscale).]

7.3 Aware and Unaware Drones

[Limits table; best ring under each threat; figures/q4_downtime_limits.png (optional).]

7.4 How Many Sensors Make Up for Downtime?

[Redundancy rule table (Da = 1, 3, 10); FIGURE: figures/q4_redundancy.png (grayscale).]

8 REAL-WORLD INTERPRETATION

8.1 From Physical Units to Pe and Da

[Mapping v, rho, D (OU maneuver model), lam0 to Pe and Da; sourced vs assumed inputs.]

8.2 Scenarios

[TABLE: results/real_world_mapping.md (nine scenarios).]

8.3 Sensitivity to Assumed Inputs

[TABLE: results/real_world_sensitivity.md.]

8.4 Design Rules

[Short list of operational rules supported by the results.]

9 DISCUSSION

9.1 Implications for Counter-UAS Planning

[What a planner should do differently.]

9.2 Limitations

[Model assumptions: isotropic diffusion, radial drift, independent sensors, Gaussian footprints, 2-D, single drone, assumed detection times; global optimality not proved for K = 6.]

9.3 Future Work

[Swarms; adaptive/informed drones; terrain; K = 6 certificate; field data.]

10 CONCLUSION

[Conclusion.]

NOTES

[Parenthetical notes marked with superscript letters (MOR does not allow footnotes).]

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[Acknowledgments; AI-assistance disclosure [TODO: confirm MOR policy]; data/code availability (GitHub: github.com/erasmorivas3540).]

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A PROOFS

[Longer proofs moved from Section 6 (from paper/theory_section.tex).]

B BRANCH-AND-BOUND DETAILS

[Bounds, error control, run statistics (experiments/bnb_straight.py, results/bnb_K4_*.json).]

C ACRONYMS

[Acronym list (optional).]